Mentl: a language whose compiler proves what software will never do

Hard Tech Week 2026 · Nucleus × Startup State open call · Morgan Espitia · sonofagoose@pm.me · github.com/ampactor-labs/mentl

THE PROBLEM

AI now helps write the software that runs machines: batteries, flywheels, flight controls. People test that code and review it, and then they trust it. No mainstream language lets a team write "this never touches the internet" or "this never writes a file" and have the compiler prove it.

WHAT EXISTS

Mentl is a programming language I have built since March 15, 2026: 2,610 commits before this week, open source. You write a rule into the code, and the compiler checks it through every function the code calls. Break the rule and the program won't build. The compiler is written in Mentl and compiles itself. It runs in a browser at space.ampactor.dev, and ampactor.dev/wasatch plays a piece of music, written from Salt Lake weather readings, whose per-sample code it proves never allocates memory or reaches the network.

sonido is my audio library, built since January 25, 2026: effects that run in plugins, in a browser and on a guitar pedal.

BUILT THIS WEEK

- A guitar pickup picks up the power grid's hum in a room's wiring, and sonido measures its frequency to a thousandth of a hertz and shows it on sonido's tuner.
- A model of a Torus Station decides, typically in under a millisecond, what its flywheel and battery should do: ride out tomorrow's rush hour, catch a real 2019 grid failure, keep the building on when the grid goes dark.
- That decision is written in Mentl and proven unable to reach the internet. Add one line that reads a price online and the compiler refuses to build it.

WHO NEEDS IT IN UTAH

- **Torus**, Salt Lake City: its Stations steady the grid, and the software inside each one is where a proof would matter first. Torus is judging this demo. The first conversation I'm asking for: Torus engineering, this month.
- **Hill AFB, 309th Software Engineering Group**: maintains flight software for the F-16, the F-35 and the missile fleet, where every claim about the software needs evidence.

NINETY DAYS

Day 30	Form the Utah company and register for federal grants, which takes weeks. Run the controller on a battery inverter on my bench.
Day 60	Run it beside a real battery at one Utah site, alongside the site's own controls.
Day 90	A pilot agreement with that site, and an SBIR proposal drafted with Nucleus Grow.

THE $10,000

$7,000	a battery inverter and battery for the bench, so the controller drives real power hardware
$3,000	forming the company, federal grant registration, and installing at the pilot site

WHAT I NEED

- A Utah battery or flywheel operator who will host the pilot.
- An introduction to the 309th, through 47G, and Nucleus Grow's help with the SBIR proposal.
- A co-founder who knows power electronics.

The first business is paid pilots with operators of batteries and flywheels. The second is a hosted check that stops AI-written control code that breaks a rule from merging. The Utah company is not formed yet; that comes first. The demo: ampactor.dev/hum, and its source at github.com/ampactor-labs/hum. Morgan Espitia has built Mentl since March and sonido since January, alone, in Salt Lake City, and wrote this week's controller in the language.